Extraction of genomic DNA from tissue culture cells

Risk assessment

– Work with human-derived material or transgenic cell lines (BSL-2)

▷ Wear gloves, safety glasses, lab coat

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This is a bench card. Full protocol available online.

›› *Preparation of cell lysates*

(1.) Prepare 10 µL proteinase K in a sterile microcentrifuge tube.

(2.) Harvest 1×10^6–5×10^6 cultured cells by trypsinization or equivalent method. Wash pellet in PBS. 🕮

(3.) Resuspend pellet in 200 µL PBS and transfer to tube containing proteinase K.

(4.) *Optional:* Add 4 µL RNase A. Mix briefly. Incubate 2 min at room temperature.

(5.) Add 200 µL cell lysis buffer. Vortex to obtain a homogenous solution.

(6.) Incubate at 55 °C for 10 min to complete protein digestion.

(7.) Add 200 µL Buffer PB. Mix thoroughly.

›› *DNA purification*

(1.) Apply the lysate on a silica membrane spin column. Centrifuge at 11 000 × *g* for 1 min or use a vacuum manifold.

(2.) Wash the silica membrane with 650 µL Buffer PE.

(3.) Repeat the wash step once more.

(4.) Dry membrane by centrifugation at 11 000 × *g* for 2 min.

(5.) Place the spin column into a clean microcentrifuge tube.

(6.) Apply 25–200 µL 5 mM Tris pH 8.0. Incubate 1 min. Elute by centrifugation at 11 000 × *g* for 1 min. 🐞

Storage of genomic DNA

(1.) Store at 4 °C or at room temperature for up to six months under slightly basic conditions (pH 8.0) or at −20 °C for long-term storage. 🕮

Quality assurance: Check DNA concentration after storage at 4 °C or room temperature to compensate for evaporation. 💎

🖉 Recipe (available online) 🐞 Troubleshooting (available online) 🕮 Notes (available online)

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